



READING ACTIVITY

"ImagIngDev: A New Approach for Developing Automatic Cross-Platform Mobile Applications Using Image Processing Techniques"

The aim of this work is propose and describe ImagIngDev, a new approach for developing automatic cross-platform mobile applications using image processing techniques. As proof of concept, we presented ImagIng Tool and compared its performance with respect to similar cross-platform application development tools. Our main contribution to software development is ImagingDev, a new, intuitive and agile approach aimed at novice developers for automatically developing cross platform mobile applications. Our proof of concept, ImagIng Tool, can generate mobile applications on four different software platforms: Android™, Windows Phone™, iOS™ and FirefoxOS™. Also, it relies on image processing techniques to recognize user interface design patterns (UIDPs) inside user interfaces, and generates the source code for cross-platform and multi device applications. As proof of concept, we developed ImagIng Tool for automatic generation of cross-platform mobile applications by using image processing techniques. Results from the evaluation demonstrated that ImagIng Tool has attractive benefits if compared to similar cross platform application development tools. Such benefits include higher learnability and usability. ImagIngDev and ImagIng Tool can effectively solve current problems in mobile applications development, such as automatic code generation for cross-platform mobile applications, since they allow users to generate cross-platform and multi-device mobile applications in an easy and intuitive way.

Reference(s)

Rosales-Morales, V. Y., Sánchez-Morales, L. N., Alor-Hernández, G., García-Alcaraz, J. L., Sánchez-Cervantes, J. L., & Rodríguez-Mazahua, L. (2019). IMAgingDev: A new approach for developing Automatic Cross-Platform mobile applications using image processing techniques. *The Computer Journal*, 63(5), 732-757. <https://doi.org/10.1093/comjnl/bxz029>

Submission Note (Individual Activity Submission)

Use the Activity template attached in MS Teams Assignment.

Submit a PRINTED document and upload the Softcopy (PDF file) in MS Team

Understanding the Content

What are the main ideas or arguments presented in the journal? Discuss thoroughly.

The article talks about ImagIngDev, a tool that can turn a drawing of an app screen into app code. It can make code for different kinds of phones, so a beginner does not have to start each version from zero. The authors tested it and found that it was easy to learn and finished their sample app in fewer steps than another tool.



Critical Analysis

Do you agree or disagree with the author's perspective? Why?

I agree with the idea because drawing a screen first would help me plan what I want to make. In the article's test, ImagIng Tool took about four minutes while the other tool took about six, but the test did not include fixing the app later. I would still check the buttons and code because a drawing alone may not show everything the app needs.

Application and Relevance

How does the reading relate to your own experiences, studies, or field of interest?

Can the ideas or findings in the article be applied to real-world problems? If so, how?

How might this reading influence your future learning or professional practice?

This connects to my Flutter class because I also make screens and think about how an app will look on different phones. A small shop could draw a simple product page and use a tool like this to make a first version of its app. I want to learn how to improve the code it makes so the app works well for real users.

Personal Reflection and Further Exploration

What did you learn about yourself or your own thinking from engaging with this reading?

What questions do you still have after reading the article?

What additional research would you like to do based on this reading?

This reading made me see that planning an app with a drawing can make coding feel less hard. I still wonder if the tool can understand a messy drawing or a screen with many details. I would like to try it with one of my own designs and compare it with the same screen I build in Flutter.